

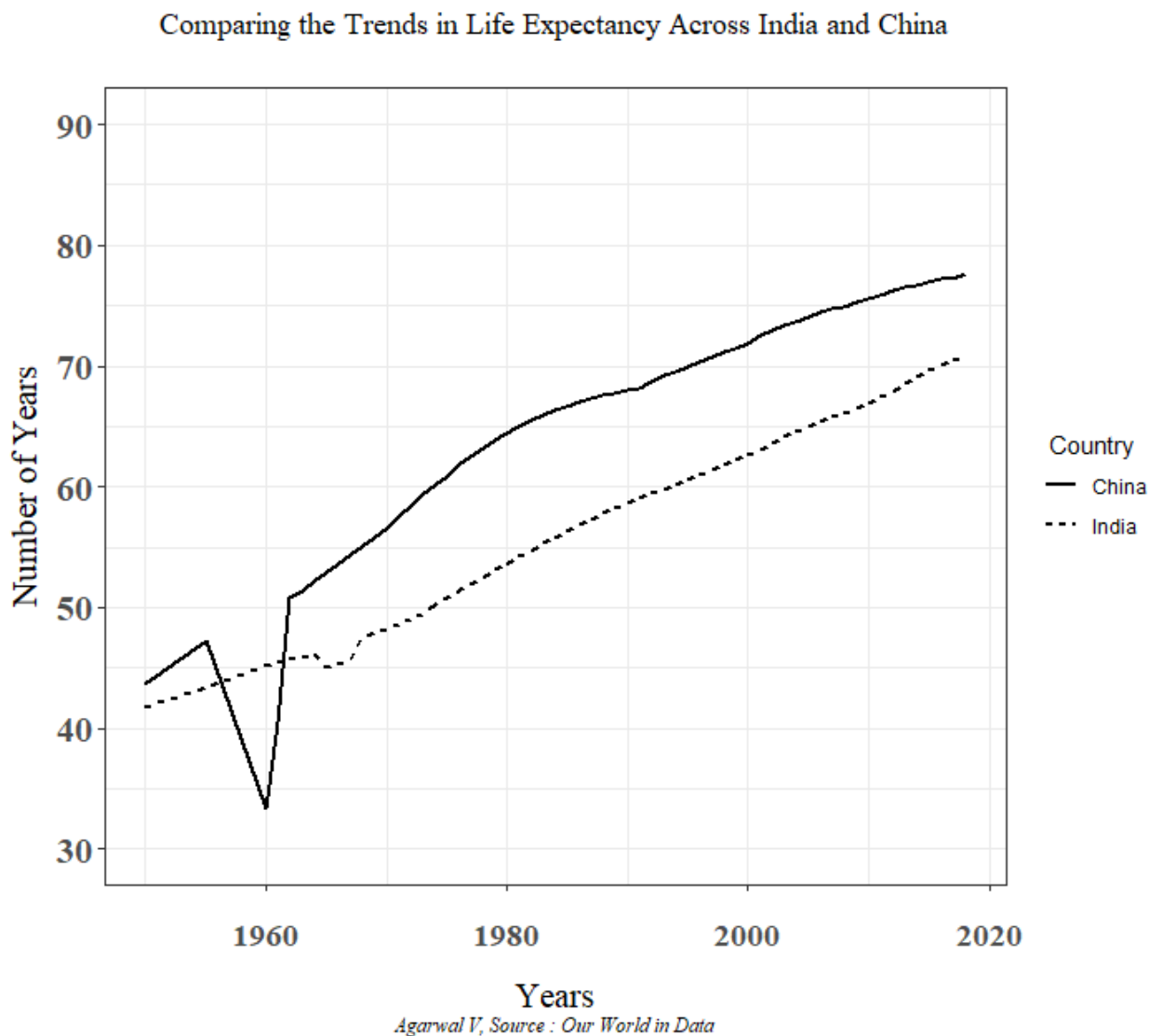
Indian and Chinese Economic
Development in Comparative
Perspective.

Data Assignment 1

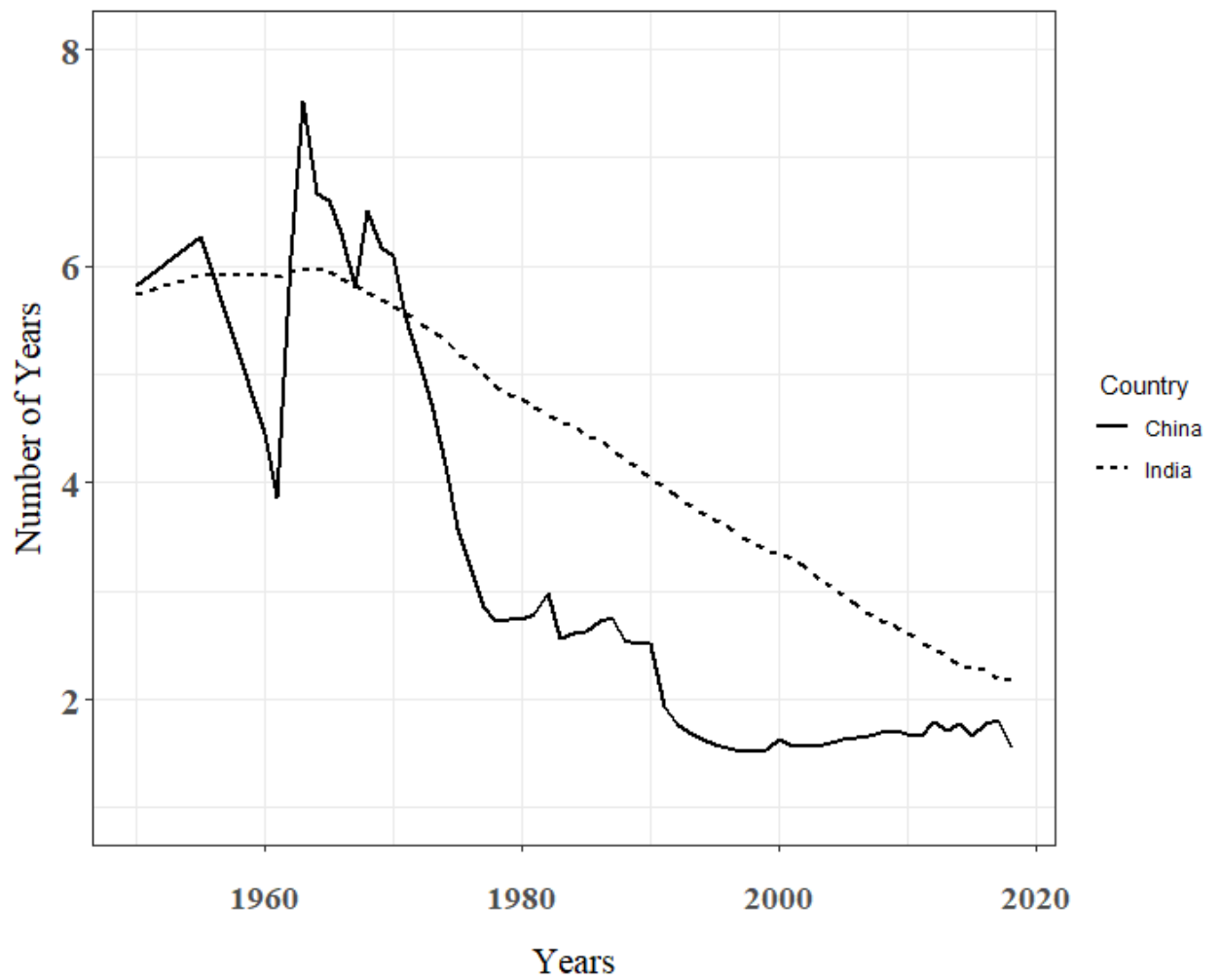
Vaibhav Agarwal

5th September, 2023

In three separate charts, plot life expectancy, fertility rate and years of schooling in India and China over time. Indian and Chinese trends should be on the same graph for ease of comparison.

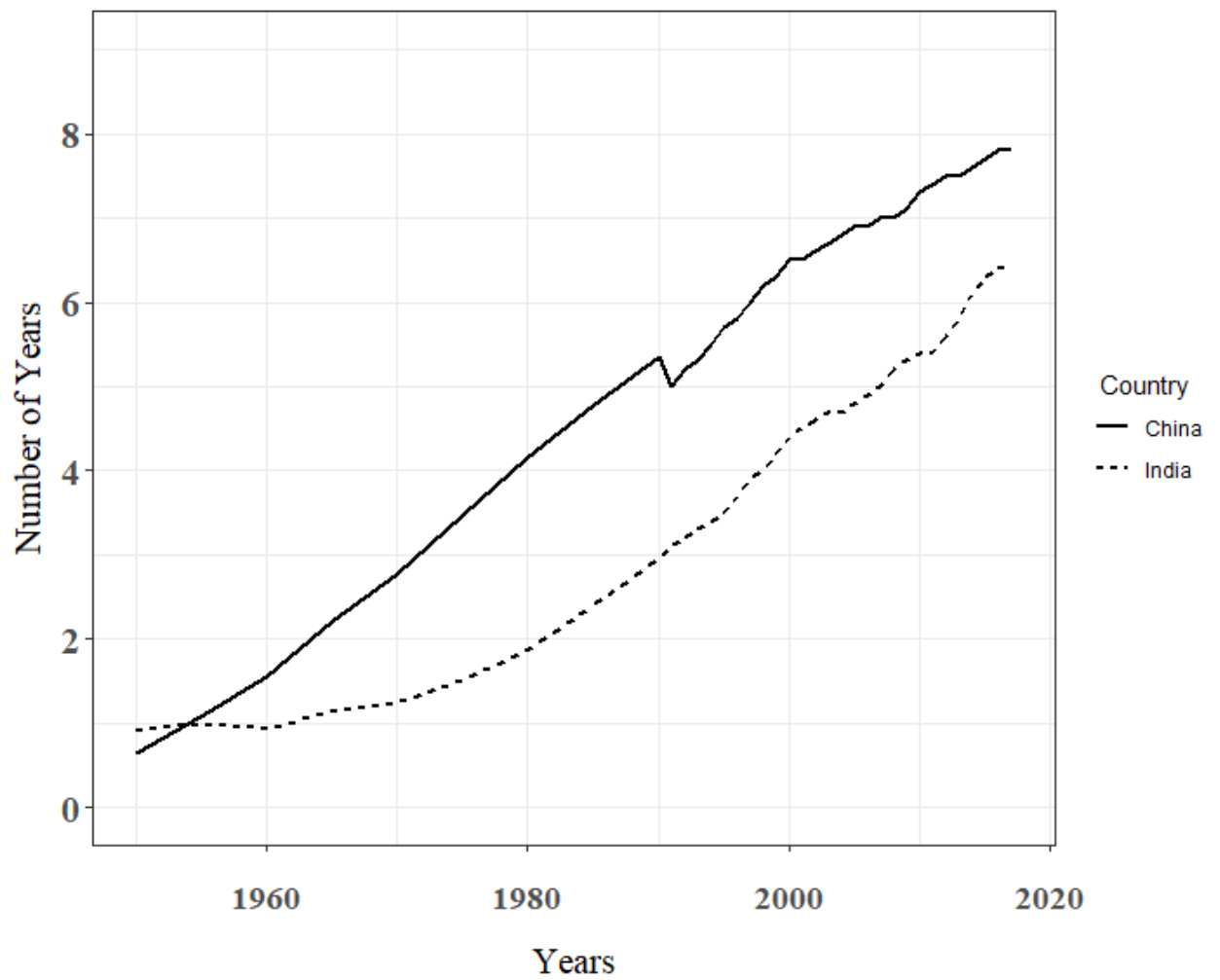


Comparing the Trends in Fertility Rates Across India and China



Agarwal V, Source : Our World in Data

Comparing the Trends in Years of Schooling Across India and China



Agarwal V, Source : Our World in Data

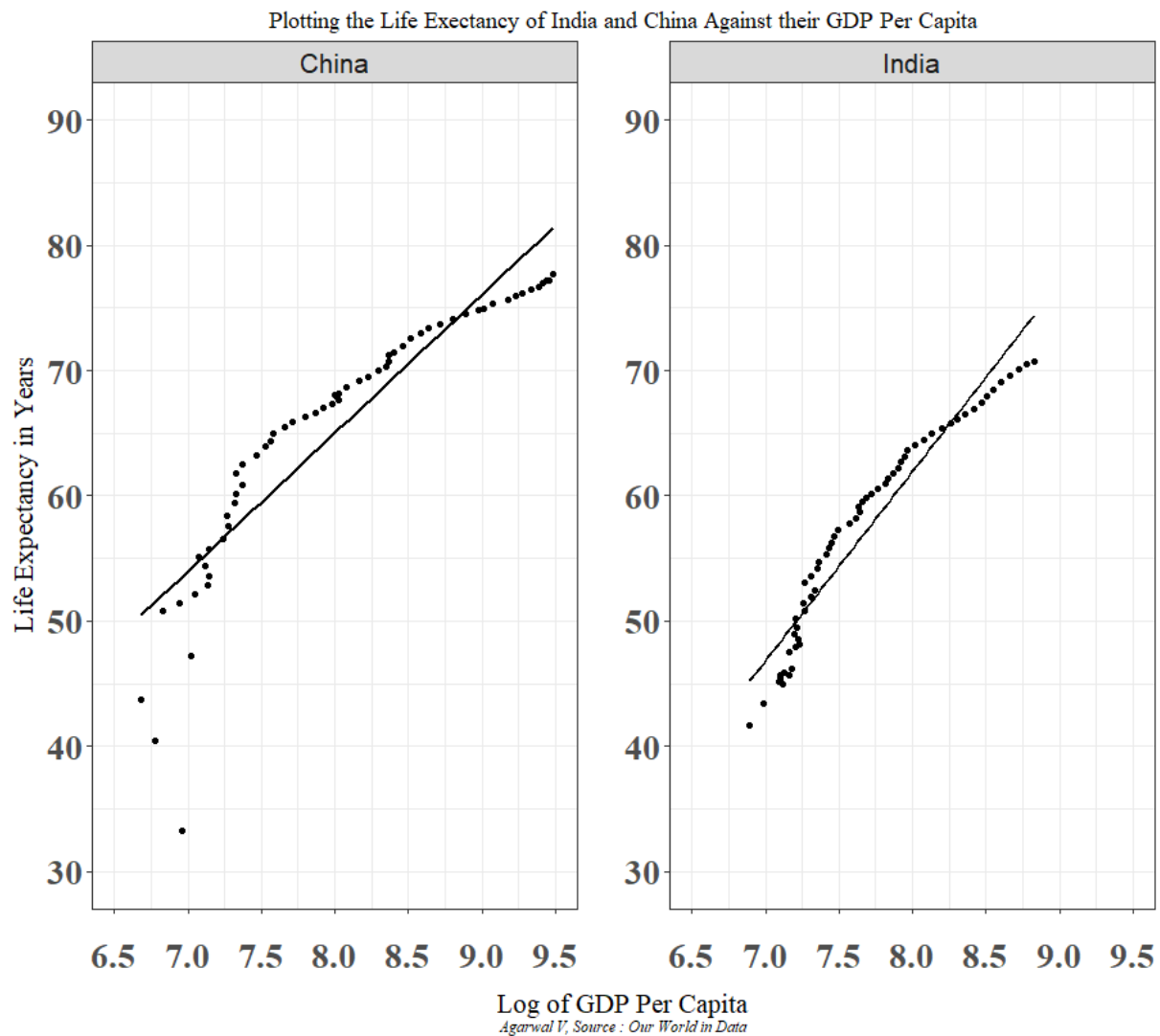
Write a brief note (approx 250 words) reflecting on the trends you observe.

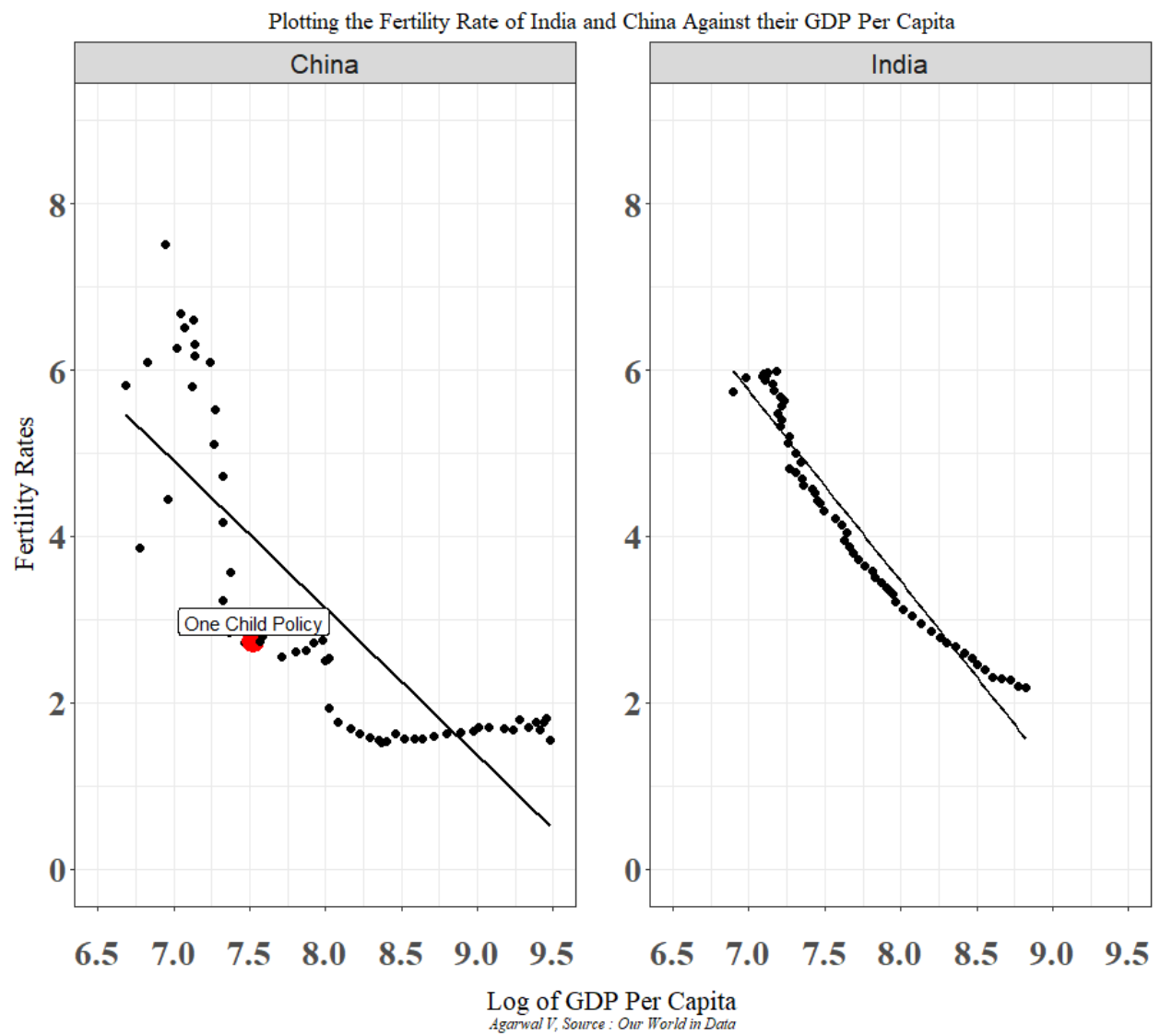
Post independence India and post revolution China were two large emerging economies of Asia. Both economies were predominantly agrarian with similar human development indicators; It is quite surprising to see India had higher levels of schooling than China in the 1950's. Fast forward decades later, China has surpassed India with higher human and economic development indicators. Post the 1970's; life expectancy, fertility rates and years of schooling; witnessed a rise at an increasing rate. The gap between India and China had peaked during the late 1980's; present day India and China are far apart, however, we can observe convergence between the two nations in the recent decade.

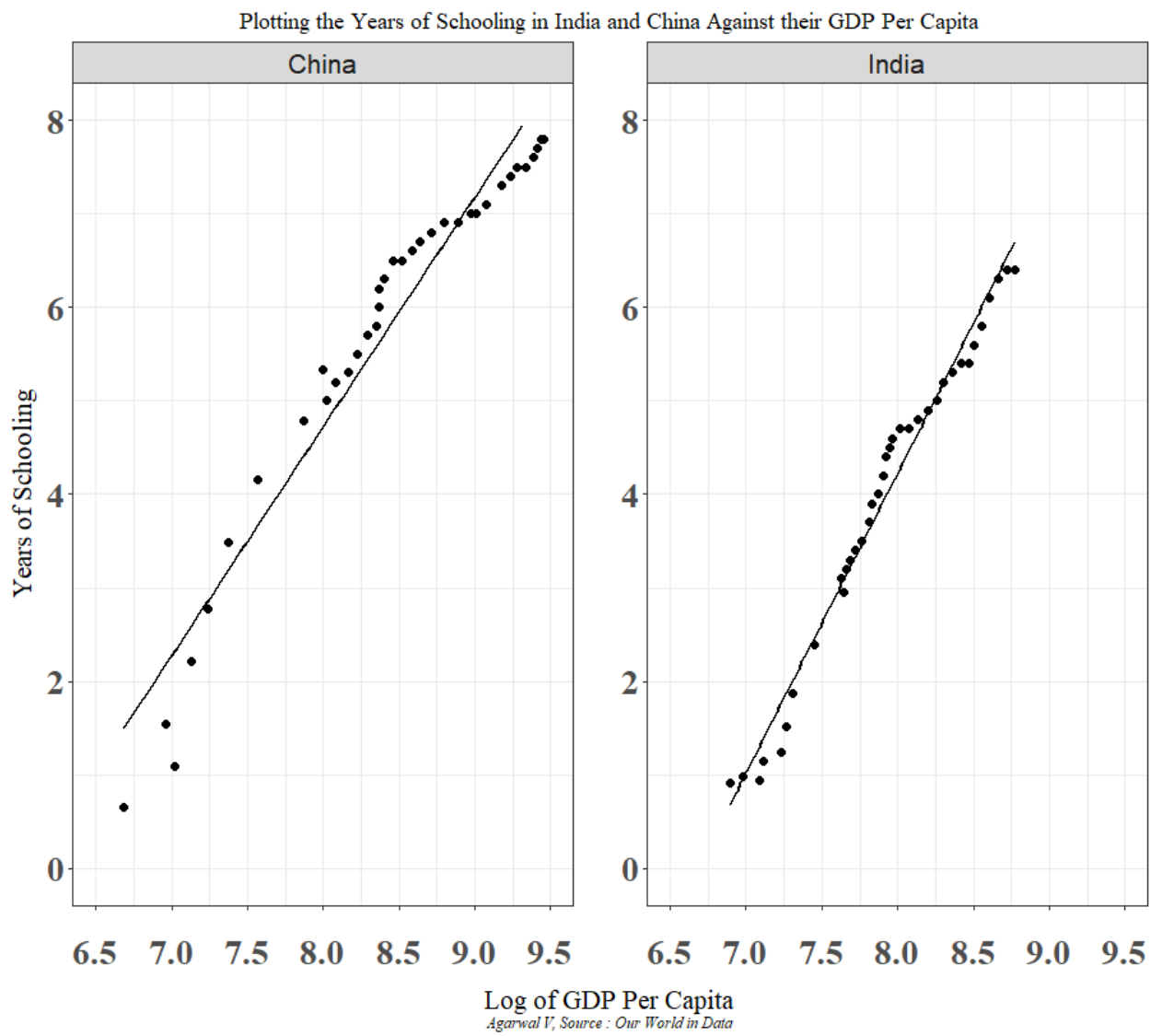
The major dip in life expectancy and fertility rates can be observed in 1960; China has witnessed a major famine during the 1960's which caused a drop in the life expectancy as well as reduced births. Post the 1960's China journey has been commendable, flying ahead of India by a big margin in less than two decades post revolution.

The effects of the one child policy which was introduced in 1979 can be seen starting in the early 1990's and achieving fertility rates below 2. India has managed to reach a fertility rate of 2 or the replacement rate, just recently. Another interesting observation from the data is the stagnation of years of schooling during the mid 1950's to early 1960's. This can be connected to the 2nd five year plan which focused on rapid industrialization as well as Nehru's vision of constructing elite educational institutions.

In separate charts, plot life expectancy, fertility rate and years of schooling in India and China as a function of log GDP per capita. Here you will have six separate graphs. & Fit a linear trend to the above plots to quantify the relationship between GDP growth and the three outcome variables.







Prepare a table that lists all six coefficients and interpret the coefficients you got

Estimating the Impact of GDP Per Capita on Human Development Indicator						
Dependent variable:						
	Life Exp. India (1)	Life Exp. China (2)	Fertility R. India (3)	Fertility R. China (4)	Years of Schooling India (5)	Indian of School China (6)
Log of GDP Per Capita (India)	15.100 ^{***} (0.552)		-2.292 ^{***} (0.077)		3.205 ^{***} (0.081)	
Log of GDP Per Capita (India)		11.046 ^{***} (0.652)		-1.768 ^{***} (0.167)		2.451 ^{***} (0.103)
Constant	-58.864 ^{***} (4.249)	-23.344 ^{***} (5.261)	21.797 ^{***} (0.591)	17.283 ^{***} (1.346)	-21.409 ^{***} (0.643)	-14.882 ^{***} (0.866)
Observations	61	61	61	61	36	36
R ²	0.927	0.829	0.938	0.656	0.979	0.944
Adjusted R ²		0.827	0.937	0.650	0.978	0.942
Residual Std. Error	2.270 (df = 59)	4.178 (df = 59)	0.316 (df = 59)	1.069 (df = 59)	0.249 (df = 34)	0.474 (df = 34)
F Statistic	748.043 ^{***} (df = 1; 59)	286.924 ^{***} (df = 1; 59)	890.150 ^{***} (df = 1; 59)	112.356 ^{***} (df = 1; 59)	1,559.724 ^{***} (df = 1; 34)	569.683 ^{***} (df = 1; 34)
Note:	* p<0.1; ** p<0.05; *** p<0.01					
Table made in latex, attaching snippet						

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The regression result above indicates that the linear model is a good fit (values of R squared are closer to 1). The results are statistically significant at the 1% confidence interval. Since the model is of linear regression, we cannot observe at what point the relation between the coefficients does not hold true and begins to fall.

There exists a positive relation between an increase in GDP Per Capita, life expectancy and years of schooling. A rise in GDP per capita by 1 % will result in a rise in life expectancy by 0.15 years in India and a rise in life expectancy by 0.11 years in China; A rise in GDP per capita by 1 % will result in fall of 0.03 children per mother in India and a fall of 0.02 children per mother in China; A rise in GDP per capita by 1 % will result in a rise in years of schooling by 0.03 years in India and a rise in years of schooling by 0.02 years in China.